

An ELF file is organized into four main parts:

1. ELF Header

Contains metadata about the file, such as its type, architecture, entry point, and the locations of the program header table and section header table.

2. Program Header Table

Contains information used at runtime by the loader. It describes how segments of the file should be mapped into memory for execution (e.g., loadable segments, permissions).

3. Section Header Table

Contains a header for each section in the file. Each entry describes a section's name, type, size, and location in the file. This is mainly used by tools like `nm`, `objdump`, and `readelf`.

4. Sections

These contain the actual data of the program, such as:

- 5. `.text` → executable code
- 6. `.data` / `.bss` → variables
- 7. `.rodata` → constants
- 8. `.symtab` / `.dynsym` → symbols
- 9. `.strtab` / `.dynstr` → symbol names

In short:

- Headers describe the structure
- Sections contain the real content